Bug Fixes for Geometric Unity: A Charitable Formalization in Lean

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Abstract

We formalize in Lean 4 with Mathlib as much of E. Weinstein's *Geometric Unity: Author's Working Draft v1.0* as is stated precisely enough to formalize, reading the draft as charitably as we can. The finite-dimensional claims of Sections 3, 4 and 8 are correct. A decomposition that the draft asks to have checked is consistent. Sections 6 and 7 contain a sign error that makes two lemmas false as printed, and no single sign convention makes the printed lemmas simultaneously true; we give a three-sign repair under which all of them hold. After the repair, and granting every standard fact the proof assistant's library cannot yet express, one thing remains: the field equation is built from an operator the draft does not define. We then try to supply the missing definition from the literature. With the textbook choice of forms, the draft's sample operator is the Ricci tensor if read as printed, and is exactly the Einstein tensor, with the draft's coefficient, under one of eight readings of its brackets. So the part of the operator that can be reconstructed is the part that was already known, and the part that would be new is the part that is missing.

1 Ground rules

The draft [1] says of itself that "the interested community is welcome to send bug fixes" (footnote 9). This note takes that literally. A proof assistant is a good tool for the job because it has no opinion of the author. It accepts what is defined and proved and stops at what is not.

The draft also says, in the footnote to its title, that its author "is not a physicist and is no longer an active academician, but is an Entertainer", and it calls itself "this work of entertainment". We do not take that literally. The document states definitions, lemmas and proofs, and it is offered in public as a theory of physics, so we check it as mathematics. A lemma is true or false whatever its author calls the document it appears in. The same footnote asks that the work not be built upon without permission; this note builds nothing on it and only checks what is there.

We read the draft under four rules, each chosen in its favor.

1. **Standard mathematics is granted.** Mathlib has no spin structures, no connections on associated bundles, no Hodge star on bundle-valued forms. Wherever the draft uses such

*Claude is an AI model made by Anthropic. This note and the accompanying Lean code were written by Claude at the request of a user, who chose the topic and asked that it appear under this name. It is not a statement by Anthropic. Every mathematical claim marked as checked is verified by the Lean kernel and does not depend on trusting the author.

things in the standard way, we assume they exist and behave as the textbooks say. None of Mathlib's gaps is counted against the draft.

2. **Conflicting conventions are resolved in the draft's favor.** Where the text is inconsistent we look for the reading that makes the most statements true.

3. **What can be repaired is repaired.** An error with a fix is reported with the fix.

4. **No physics is judged.** We ask only whether the mathematics is defined and whether the stated lemmas follow. Objections concerning anomalies, unitarity or supersymmetry [2] are outside what a proof assistant checks, and we take no position on them.

Sections 5–7 of the draft are algebra on three pieces of data, and the draft's own proofs use nothing else: a group $\mathcal{H}$ (the gauge group), an abelian group $\mathcal{N}$ on which it acts (the ad-valued one-forms, with $h \cdot n = hnh^{-1}$), and a map $c : \mathcal{H} \to \mathcal{N}$, $c(h) = h^{-1}(d_{A_0}h)$, satisfying the Leibniz rule $c(h_1h_2) = h_2^{-1}c(h_1)h_2 + c(h_2)$. We formalize at that level of generality. A statement proved from these axioms holds in the draft's geometric setting. A statement that fails from these axioms fails there too whenever the explicit error term is nonzero.

2 What checks

Table 1 lists every statement examined. Everything before Section 6 that could be checked is correct.

Draft	Statement	Result
§3.1, (3.8)–(3.9)	Signature of the Frobenius form on the fibre of metrics	True
§3.5	Signature totals leading to $(7, 7)$	True
(3.21)–(3.27)	Clifford degrees forming $gl(64, \mathbb{R})$, $so(64, 64)$, $u(64, 64)$	Consistent
(4.2)–(4.3)	Internal spinor dimensions 32 and 16	True
(8.2)–(8.6)	$2^{14} = 128^2$ and the degree split of $u(64, 64)$	True
Def. 5.1, (5.9)	The inhomogeneous gauge group $\mathcal{G}$ is a group	True
(6.2)	Right action of $\mathcal{G}$ on connections	True
Lemma 6.1	The tilted map τ is a homomorphism	True
§6.2	$\hat{\tau}$ parameterizes the stabilizer of A_0	True
Lemma 6.3	π is constant on tilted orbits	True as printed
Lemma 6.4	Left action of $\mathcal{G}$ on $\mathcal{N}$	True
Lemma 6.5	The bi-connection map is equivariant	**False as printed**
Lemma 7.2	The augmented torsion is equivariant	**False as printed**
§§6–7	Lemmas 6.3 and 7.2 under one sign of τ	**Inconsistent**
Def. 8.1	The forms Φ_i	**Not determined**
(9.2)	The shiab operator	**Not defined**
(8.1)	The conjugation recipe is equivariant for every Φ	True
(9.3)	The sample operator is Einstein's contraction	One reading of eight
(9.8)	Gauge covariance of the field equation	Conditional

Table 1: Statements examined. All results are machine-checked except the row for (9.3), which is numerical, and where Section 8 says otherwise.

Two of these deserve a remark.

The signature. For a metric of signature $(1,3)$ the form (3.8), $\langle A, B\rangle = \mathrm{Tr}(g^{-1}Ag^{-1}B)$, is diagonal on symmetric matrices with seven coefficients of one sign and three of the other, and the trace direction lies among the seven. So (3.8) as defined has signature $(7,3)$, and the $(4,6)$ the draft uses comes from reversing the sign of the trace direction by hand. The draft says this is a choice (footnote 2), so we record it as true.

A request fulfilled. Footnote 3 says of the decomposition (3.27) of $u(64,64)$ into exterior powers that it "was taken from a decades old file and should be checked by someone more current". We checked it. Given the standard rule for the degrees appearing in a Clifford commutator, the degrees congruent to $1, 2 \bmod 4$ are closed under the bracket, the degrees congruent to $0, 3 \bmod 4$ twisted by i bracket with them and with each other as (3.27) requires, and the dimensions are $8128 + 8256 = 128^2$. The decomposition is consistent.

3 The bug

The draft defines the tilted map in (6.4) by

$$\tau(h) = \big(h,\ -h^{-1}(d_{A_0}h)\big) = (h, -c(h)),$$

and the augmented torsion in (7.3) by $T_{(\varepsilon,\varpi)} = \varpi - c(\varepsilon)$. Lemma 7.2 asserts $T_{g\cdot\tau(h)} = h^{-1}T_g h$. Using the group law (5.9) and the Leibniz rule,

$$T_{g\cdot\tau(h)} = \big(h^{-1}\varpi h - c(h)\big) - \big(h^{-1}c(\varepsilon)h + c(h)\big) = h^{-1}T_g h \ -\ 2\,h^{-1}(d_{A_0}h).$$

The draft's proof (7.4) writes $-h^{-1}(d_{A_0}h)$ in its first line and $+h^{-1}(d_{A_0}h)$ in its second. The same change of sign occurs between the lines of (6.20), the proof of Lemma 6.5.

Theorem 1 (`torsion_mul_tauMinus`, `biConn_mul_tauMinus`)**.** *With τ as printed in* (6.4)*, Lemma 7.2 fails by $-2c(h)$, and the first component of Lemma 6.5 fails by the same term.*

The charitable response is that the sign in (6.4) is a misprint. With $\tau(h) = (h, +c(h))$, which is also the form in the 2020 lecture as transcribed in [2], Lemmas 6.5 and 7.2 are true. But Lemma 6.3, with π as printed in (6.11), then fails:

$$\pi\big(g\cdot\tau(h)\big) = \pi(g) + 2\,\varepsilon\,(d_{A_0}h)\,h^{-1}\varepsilon^{-1}.$$

Theorem 2 (`lemma63_and_lemma72_incompatible`)**.** *Suppose $\mathcal{N}$ has no 2-torsion. Fix either sign for τ. If Lemma 6.3 and Lemma 7.2 both hold as printed for that sign, then $h^{-1}(d_{A_0}h) = 0$ for every $h \in \mathcal{H}$.*

So the printed text cannot be made true by choosing a convention. The draft anticipates something of the kind: footnote 7 says there "may be 2 or more conflicting conventions in this section". The theorem says where they are and that they cannot be reconciled without changing a definition.

4 The fix

Change three signs and leave everything else as printed:

$$\begin{aligned} (6.4)\quad & \tau(h) = \big(h,\ +h^{-1}(d_{A_0}h)\big), \\ (6.11)\quad & \pi(\varepsilon,\varpi) = \varepsilon\varpi\varepsilon^{-1} - (d_{A_0}\varepsilon)\varepsilon^{-1}, \\ (6.15)\quad & (\varepsilon,\varpi)\cdot\gamma = \varepsilon(\varpi+\gamma)\varepsilon^{-1} - (d_{A_0}\varepsilon)\varepsilon^{-1}. \end{aligned}$$

Theorem 3 (`GU.Repaired`)**.** *With these three changes, Lemmas 6.1, 6.3, 6.4, 6.5 and 7.2 all hold, the stabilizer of A_0 is the image of the map $\hat{\tau}$ of* (6.9) *as printed, the action* (6.15) *is the one induced by π from left multiplication, and right multiplication by $\tau(h)$ acts on $A_0 + \varpi$ as the gauge transformation h.*

The repair also resolves an oddity in the printed text, where the "tilted" map τ of (6.4) and the "flipped" map $\hat{\tau}$ of (6.9) are given by the same formula. After the repair they are different maps with different jobs, which is evidently what was meant.

We regard Sections 5–7, so repaired, as correct. From here on we use the repaired version.

5 What cannot be fixed

The first-order field equation (9.8) is

$$\circledcirc_\varepsilon F_{A_\omega} \ = \ -\kappa_1 * T_\omega.$$

The curvature F and the Hodge star are standard and we grant them. T is the torsion of Section 7, now verified. The remaining ingredient is the shiab operator $\circledcirc_\varepsilon : \Omega^2(Y,\mathrm{ad}) \to \Omega^{d-1}(Y,\mathrm{ad})$. It is the one new operator in the equation, and the draft says the following about it.

- "Then a Shiab contraction operator might look like: [(8.1)] or perhaps a sum of such terms."
- "The author is no longer in a position to go chasing after the complete picture and simply details some of the available tools for customizing such operators."
- "The particulars of the Shiab operators are workmanlike and not of much interest."
- "Unfortunately, the author is no longer conversant in that language and has been unable to locate the notes from decades ago that originally picked out the operator of choice."
- "The Shiab operator that he settled on (but cannot yet now locate) was chosen for its properties relative to the Bianchi identity." (footnote 10)

A repair requires something to repair. Here the charitable reading has nothing to work with, because the object is absent rather than wrong.

The draft does offer one sample operator, (9.3), "which makes the parallel to Einstein's contraction of the full Riemannian curvature explicit". We tried to take that as the definition. It is built from forms Φ_1 and Φ_2, and Definition 8.1 says only that the Φ_i are "a basis for the invariant subspaces of" $[\Lambda^i(\mathbb{R}^{7,7}) \otimes u(64,64)]^{Spin(7,7)}$. This does not determine them.

Proposition 4 (`equivariant_mul_commuting`)**.** *Let a group act on a ring A by conjugation and on a set V. If $\Phi : V \to A$ is equivariant and $\lambda \in A$ commutes with the group, then $v \mapsto \Phi(v)\lambda$ is equivariant.*

In $Cl(7,7)$ the volume element λ commutes with $Spin(7,7)$. Taking $\Phi(v) = \gamma(v)$ gives a second invariant $\gamma(v)\lambda$, of Clifford degree 13 where the first has degree 1. The draft's own equation (8.5) places both degrees inside $so(64,64) \subset u(64,64)$. The same holds for Φ_2 with degrees 2 and 12, by (8.5) and (8.6). So each space the Φ_i are drawn from is at least two-dimensional, and the sample operator (9.3) has free parameters that the draft does not fix, in addition to the constant κ_1, which is given no value.

What can be stated in Lean is therefore this.

Theorem 5 (`upsilon_equivariant`)**.** *Let $\circledcirc$ be* any *family of additive maps $\circledcirc_\varepsilon : \Omega^2 \to \Omega^{d-1}$ satisfying $\circledcirc_{\varepsilon h}(h^{-1}\xi h) = h^{-1}(\circledcirc_\varepsilon \xi)h$. Then, granting the standard equivariance of curvature and of the Hodge star, $\Upsilon_g = \circledcirc_\varepsilon F_g + \kappa_1 * T_g$ satisfies $\Upsilon_{g\cdot\tau(h)} = h^{-1}\Upsilon_g h$.*

This is the central claim of the theory in the strongest form we can prove: the field equation is gauge covariant for every operator with the stated property. It is a true theorem. It is also a theorem about a variable. The draft supplies no value for the variable, and the claims that depend on its particular value, including that it annihilates the Weyl curvature, that the equation reproduces Einstein's, and everything in Sections 10–12, cannot be stated.

6 Filling the gap from the literature

Rule 3 says to repair what can be repaired, so we tried to supply the operator ourselves.

Equivariance is free. The recipe of (8.1) conjugates a fixed form by the group element: $\circledcirc_\varepsilon \xi = [\varepsilon^{-1}\Phi\varepsilon \wedge *\xi]$.

Proposition 6 (`conjugated_pairing_equivariant`)**.** *For any fixed Φ and any equivariant pairing B, the operator $\circledcirc_\varepsilon \xi = B(\varepsilon^{-1}\Phi\varepsilon,\, \xi)$ satisfies $\circledcirc_{\varepsilon h}(h^{-1}\xi h) = h^{-1}(\circledcirc_\varepsilon \xi)h$.*

So the hypothesis of Theorem 5 holds for every operator of this shape, and the gauge covariance of the field equation is established for all of them. This is good news for the draft. It also means that gauge covariance, the property the construction is advertised for, does nothing to select Φ.

The textbook forms. The literature has a canonical candidate for the Φ_i: the tautological Clifford-valued forms

$$\Phi_1 = \sum_a e^a \otimes \gamma_a, \qquad \Phi_2 = \sum_{a<b} e^a \wedge e^b \otimes \gamma_a\gamma_b,$$

where Φ_1 is the form whose Clifford action defines the Dirac operator. We put these into the sample operator (9.3) at $\varepsilon = 1$ and applied it to the curvature of the spin connection of a metric, $\xi = \frac{1}{4}R_{efcd}\, e^e \wedge e^f \otimes \gamma^c\gamma^d$. After taking Hodge duals, (9.3) has a first term $P_f = \sum_a [\gamma_a, \xi_{af}]$, labelled "Ricci like", and a second term $Q_f = [\gamma_f, S]$ with $S = \sum_{a<b}[\gamma_a\gamma_b, \xi_{ab}]$, labelled "Ricci scalar like". The draft's notation allows each bracket to be a commutator or the symmetric product $\{a,b\} = i(ab+ba)$ of its appendix, and (9.3) prints plain brackets. We evaluated all eight combinations on random tensors with the symmetries of a Riemann tensor, in dimensions 4 and 6 with a positive-definite metric (`experiments/einstein_check.py`).

1. In every combination the Weyl part of the curvature is annihilated, as the draft asserts.

2. With commutators throughout, the second term is identically zero and the operator is the Ricci tensor alone.

3. In exactly one combination, a commutator in the first term and the symmetric product in both brackets of the second, the operator is $\mathrm{Ric} - \frac{1}{2} R\, g$, the Einstein tensor. The magnitude $\frac{1}{2}$ is the coefficient printed in (9.3). Its sign depends on conventions for the Hodge star and the Clifford relation that the draft does not state.

The reason for item 2 is short. The Riemann tensor is unchanged when its first pair of indices is exchanged with its second, which makes the two orderings in the inner bracket equal, so their difference vanishes. And had the inner bracket produced the scalar curvature, the outer bracket would be the commutator of γ_f with a scalar, which is also zero. The symmetric product adds where the commutator subtracts, and $R_{abcd}\gamma^a\gamma^b\gamma^c\gamma^d = -2R$ supplies the scalar.

Without its scalar term the equation is the one Einstein published in November 1915 and replaced later the same month, because the Ricci tensor alone is not divergence-free and the equation is then inconsistent with conservation of energy.

What the reconstruction shows. Read as generously as possible, with forms from the textbooks and the one bracket assignment that works, the draft's sample operator is correct on the gravitational part of the curvature. We are glad to be able to report that. But what it is correct *as* is the Einstein tensor in Clifford notation, which is the contraction of Einstein–Cartan gravity and has been known for a century. The curvature in the draft takes values in all of $u(64, 64)$, of which $so(7, 7) = \Lambda^2$ is 91 dimensions out of 16384. For the rest, the pair symmetry and Bianchi identity used above are not available, the literature offers no operator, and the draft does not say what $\circledcirc$ does. That remainder is where anything new in the theory would have to be.

7 What would complete it

After the reconstruction the list is short. It would suffice to state:

1. whether the forms and brackets of Section 6 are the intended ones, and which combination of each Φ_i with its volume-element twin is meant;

2. a value or a role for κ_1;

3. what $\circledcirc$ does to curvature outside $so(7, 7)$, and a proof of whatever property replaces "annihilates the Weyl curvature" there.

Items 1 and 2 are a few lines. Item 3 is the theory.

An offer. This is meant literally. The signs are fixed, the decomposition is checked, and the gravitational part of the operator is reconstructed. In the Lean project the operator is a single field, `FieldEquationInputs.shiab`, waiting for a definition. If the author can recall the operator, or choose one, it goes in that field, and the claims that depend on it become statements a proof assistant can check. We would be glad to finish the job.

8 Limits of this note

The results for Sections 5–7 are proved for the abstract data described in Section 1, which is all the draft's proofs use. The identification of that data with gauge transformations, one-forms and $h^{-1}d_{A_0}h$ is ours and is not machine-checked. The Clifford degree rule used for (3.27), the isomorphism $Cl(7,7) \cong \mathbb{R}(128)$, and the two facts about the volume element used in Section 5 are standard and are taken as known rather than derived. The bundle-level constructions of Sections 3 and 4 were not formalized; only their pointwise linear algebra was. The results of Section 6 on the Einstein tensor are numerical, for a positive-definite metric in dimensions 4 and 6 rather than signature $(7,7)$ in dimension 14, and are not formalized. Nothing here bears on whether a completed theory would describe nature.

The Lean project accompanying this note builds against Mathlib for Lean 4.35 and contains no `sorry`.

9 Conclusion, in plain terms

Here is what happened, without the notation.

The draft is a recipe. We followed it as far as it goes. The early steps are standard and correct. In the middle, two steps contradict each other because a minus sign changes partway through a calculation; we found where, and changed three signs so that everything agrees. Every long manuscript has errors of this kind and they count for little.

Then the recipe reaches its main step, the one ingredient that is new, and says that the author used to know what it was and has lost his notes. We tried the nearest thing from the standard cookbook. Used exactly as the recipe is written, it gives the version of Einstein's equation that Einstein himself discarded in 1915. Used in the one way that works, it gives Einstein's equation. Either way it gives nothing new, and it covers a small corner of what the recipe needs.

Four things seem worth drawing from this.

Wrong and undefined are different conditions. The sign errors made definite claims, so they could be tested, found false and corrected. The operator makes no definite claim, so there is nothing to test. Nguyen and Polya close their response by quoting Dyson: it is better to be wrong than to be vague. A proof assistant enforces that preference mechanically. It will take a false statement and show that it is false. It cannot do anything with a statement that has not been made.

Formalization is a test of completeness, and passing it takes no cleverness. Nothing in this note was difficult. The tool's contribution was to decline to continue past a missing definition, where a sympathetic reader continues without noticing. This is also why formalizing a theory cannot be delegated to a future, more capable tool. A better prover checks faster. It does not supply definitions on the author's behalf, and if it did, the result would be the prover's theory.

Charity is a method and not only a courtesy. We granted every standard fact, resolved every ambiguity in the draft's favor and repaired what could be repaired. What survives that treatment cannot be attributed to an unsympathetic reading. A gap that remains after the reader has done everything possible to close it belongs to the text.

Where the theory can be completed, it returns what is already known. This is the observation we would most like to be shown wrong about, and it is easy to show wrong: state the operator. Until then the claim that the equation unifies anything rests on the part of the equation that

does not yet exist. This is not evidence that the idea is false. It is the absence of anything that could be evidence either way, and the burden of supplying it rests with the person making the claim.

A last point concerns this note itself. Its author is a language model, and a language model's assurance that a proof is correct is worth nothing. That is the reason for the form the work takes. The Lean results do not ask to be believed; anyone can build the project and watch the kernel accept them. The numerical experiment is a short script and can be rerun. The same standard that this note applies to the draft is the one it should be held to.

References

[1] E. Weinstein, *Geometric Unity: Author's Working Draft, v 1.0*, April 1, 2021.

[2] T. Nguyen and T. Polya, *A Response to Geometric Unity*, February 23, 2021.